

Project Name: FIELDLINK (Offline-Mesh-Asset-Sync

Track / Theme: Disaster Management, Defense Logistics & Viksit Bharat @ 2047

APEX PLATOON

Team Members

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Imagine you are stuck in disaster and you have no internet, no way to call for help. This is the reality our field units face every day.

PROBLEM

FIELD UNITS, DISASTER RELIEF CREWS, AND CADET CONTINGENTS OPERATING IN REMOTE AREAS FREQUENTLY LOSE CELLULAR CONNECTIVITY, WHICH PREVENTS THEM FROM PERFORMING REAL-TIME INVENTORY TRACKING, MUSTER-ROLL (ROLL CALL) VERIFICATION, AND INCIDENT REPORTING.

NO SIGNALS

In remote training grounds, disaster zones, and border areas with zero cellular coverage, cloud-dependent apps become completely unusable, forcing teams back to slow, unreliable radios and paper logs exactly when speed matters most.

FRAGMENTED OPS

Without connectivity, each unit maintains its own isolated records — so when teams regroup, data is scattered, duplicated, or contradictory, and manually reconciling it afterward wastes critical time and invites human error.

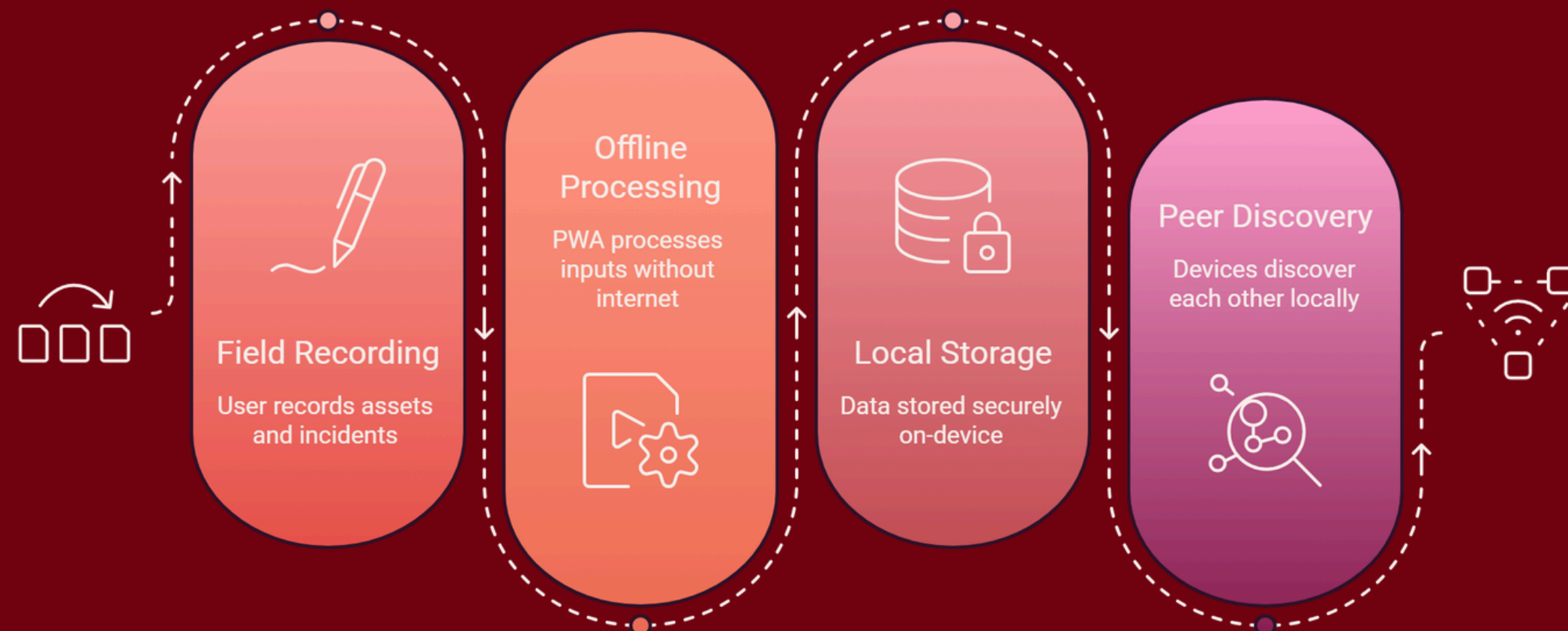
CRITICAL NEED

In disaster relief and field operations, outdated or incomplete data can cost lives — missing personnel and equipment shortages must be flagged the instant devices come into range, not hours later during a debrief back at base.

SOLUTION

An offline-first PWA that lets field personnel record assets, checklists, attendance, and incidents directly on their devices – no internet required.

Data Synchronization Process



FULLY OFFLINE

Runs entirely on-device with no cellular, Wi-Fi, or cloud needed — field teams can log data anywhere, even in the most remote or disaster-hit areas.

LOCAL STORAGE

Every entry — assets, checklists, attendance, incidents — is saved securely on-device via IndexedDB, so no data is lost even after days offline.

AUTO DISCOVERY

When devices come within Bluetooth or Wi-Fi Direct range, they auto-detect and sync in the background — no pairing, no manual action needed.

END-TO-END SYSTEM ARCHITECTURE



Asset Tracking

Track equipment deployment, status, and assignment



Peer-to-Peer Sync Layer

WebRTC · Bluetooth Low Energy (BLE) · Wi-Fi Direct



Application Layer

Asset Tracking · Muster & Roll Call · Checklists · Incident Reporting



Consistency Layer

CRDT conflict resolution · Automatic merge · Eventually consistent



Neargy Auto-sync

IndexedDB · Offline Cache · Operation Event Log



Connectivity Recovery

Cloud / Internet sync when connectivity returns

From offline data to resilient coordination

TECHNOLOGY STACK

Frontend

PWA / HTML / CSS/ TypeScript

Storage

IndexedDB · Offline Cache

Communication

WebRTC + BLE + Wi-Fi Direct

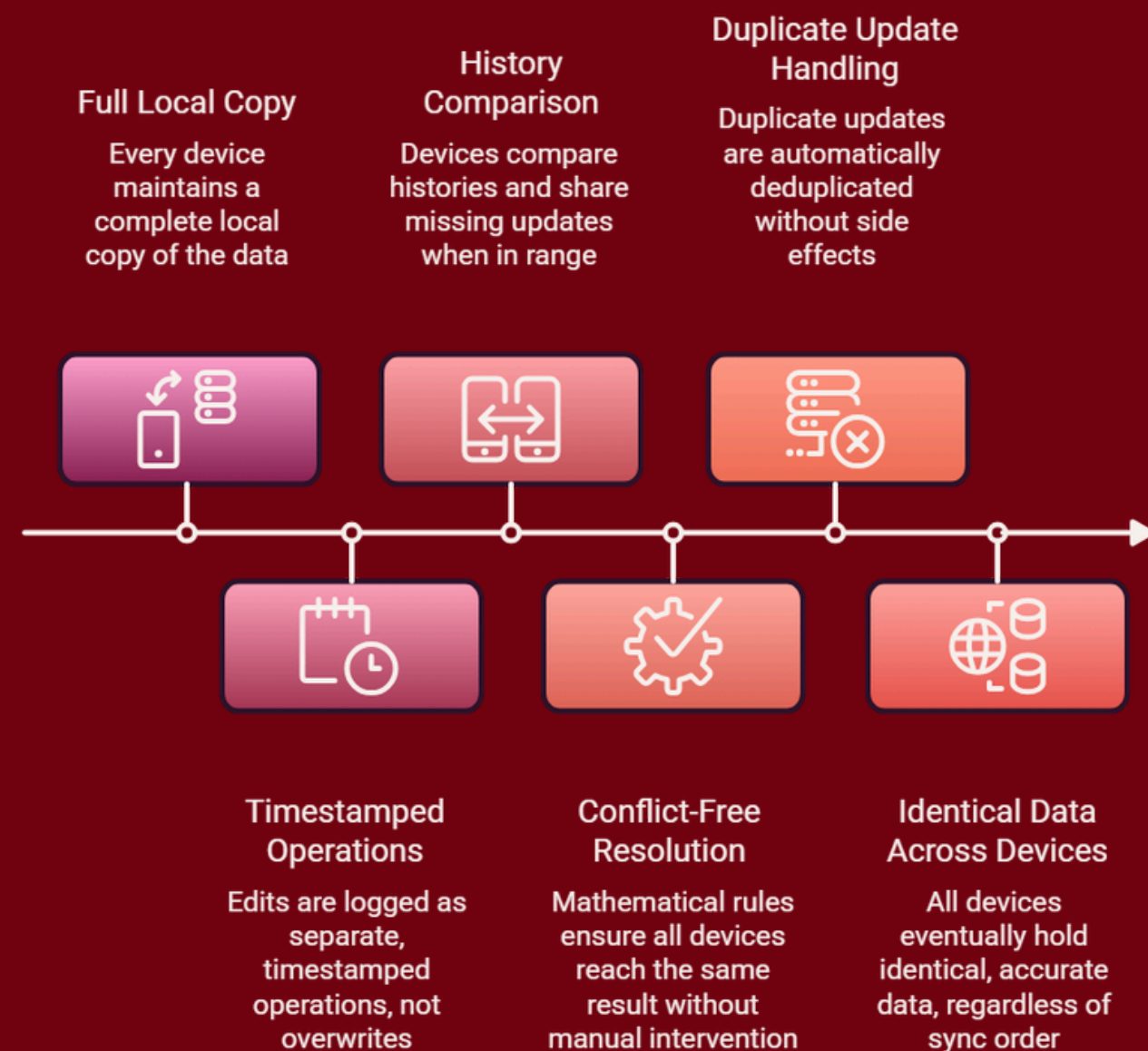
Sync Engine

WebRTC + BLE + Wi-Fi Direct

Optional Backend

Cloud sync on connectivity return

Key Features of a Decentralized Data Synchronization System



Prototype

Future Enhancements

- End-to-end encryption
- Role-based access control
- Digital signatures for records
- Smarter mesh routing
- GPS / geospatial visualization
- Multi-hop synchronization
- Analytics on reconnect

